HDMI EXTENSION BOARD – XEM8310 CONNECTIONS, POWER AND MOUNTING
MC1
BTE-040-01-F-D-A
MC2
BTE-040-01-F-D-A
MC3
QTH-040-01-F-D-DP-A
Power Jack
12 V nominal, center positive
J4
PJ-063AH
XEM8310 MODULE MOUNTING HOLES
FRONTEND 5 V POWER | CUSTOM MICRO-HDMI PORTS – NOT STANDARD HDMI
U700
TPSM63606RDLR
PCIe x4 – BRK8310 reference circuit
U2
PCA9306DCUR
J6
PCIe x8 Card-Edge Connector
U5 AP22653AW6-7
USB-C JTAG PROGRAMMER, SELECTION AND FRONTEND PORTS
USB TO JTAG PROGRAMMER
TARGET SELECTION
XEM8310 unused-pin header holes (V45; bare holes since V46)
J902
Conn_01x14
J900
Conn_02x14_Odd_Even
J901
Conn_02x14_Odd_Even
Conn_01x15
Conn_02x15_Odd_Even
Voltage test holes (V35)
J920
Voltage test holes
CUSTOM FRONTEND INTERFACE – NOT HDMI VIDEO
FRONTEND PORT 1 | JTAG SWITCH + MICRO-HDMI
FRONTEND PORT 2 | JTAG SWITCH + MICRO-HDMI
FRONTEND PORT 3 | JTAG SWITCH + MICRO-HDMI
FRONTEND PORT 4 | JTAG SWITCH + MICRO-HDMI
FRONTEND PORT 5 | JTAG SWITCH + MICRO-HDMI
FRONTEND PORT 6 | JTAG SWITCH + MICRO-HDMI
FRONTEND PORT 7 | JTAG SWITCH + MICRO-HDMI
FRONTEND PORT 8 | JTAG SWITCH + MICRO-HDMI
U611
TS3A27518EPWR

Function: U700 converts +VDCIN to FE_5V (5.02 V nominal, 6 A module rating).
PG and R703 enable all eight ports once the central rail is in regulation.

Source / changes: TI TPSM63606 SLVSGB4B Rev. B, Fig. 9-1 and Table 8-1,
https://www.ti.com/lit/ds/symlink/tpsm63606.pdf . Codex adapted prior V14 U100:
R700 40.2k, R701 10k, R702 13k (all 1%), C708 22pF set 5 V / 1 MHz; PG gates switches.
C700–C703 10uF / 50V; C704–C707 use 47uF / 16V X5R instead of prior X6S.
CBOOT/RBOOT shorted, VLDOIN tied to output. Check effective capacitance,
loop stability, input damping, startup, thermal performance and PCB layout.

Function: Connects PCIe x4 data and 100 MHz reference clock, with reset, SMBus and WAKE#. Host slot 12 V is isolated from the board 12 V input.

Source / changes: Opal Kelly BRK8310-A01 schematic, p. 2, 2022-05-19: https://pins.opalkelly.com/downloads/442/download?category_id=3
Adapted from x8 to x4: fit R28 = 0 ohm; omit R29 and unused lanes 4-7 and C21-C26/C29-C30. R2/R7 and R9-R11 retained.
Unused options R1/R8, R16/R17 and R27 are omitted.
Host +12 V goes only to TP20. Host 3.3 V and AUX power the original sideband circuitry; they do not feed the main 12 V input.
PCA9306 Rev O, Fig. 8-1, section 10: https://www.ti.com/lit/ds/symlink/pca9306.pdf ; SN74AXC1T45 Rev E: https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
AP22653 DS41186 Rev 5-2, sections 7-9: https://www.diodes.com/datasheet/download/AP22652_AP22653_AP22652A_AP22653A.pdf
Q1 Si1922EDH dual FET, product datasheet: https://www.vishay.com/en/product/67192/
V30 (Howard, 4 Nov 2026): TX AC-coupling caps C13-C20 changed from BRK8310's 0.1 µF (Kemet C0402C104K8PACTU) to 0.22 µF (Samsung CL05B224KO5NNNC, LCSC C16772, 0402 X7R 16 V); same footprint and routing.
Reason: PCIe Gen3 TX coupling range 176-265 nF (commonly cited PCIe Base Spec value, also in AMD UG583; the PCI-SIG document itself was not read); 220 nF also covers Gen1/2. C7/C8 (REFCLK) unchanged.
Working target: VIO1 = 1.8 V (Howard, 3 Nov 2026; needed for PCA9306 >= 1.2 V and Bank 64 LVDS); set in XEM before testing and verify all Bank 64 I/O. Factory default 1.0 V is below PCA9306 range: https://docs.opalkelly.com/xem8310/device-settings/
ERC power flags at U2/U6 VREF2 represent their R13/R23 200k resistor-fed bias, not independent regulators. Confirm bias and startup.
PCIe/HDMI combined Vivado design, clock, VIO1 choice, host sequencing and signal integrity remain to be verified.

Function: HDMI1 selects shared JTAG and VREF.
100R series parts alone do not provide 5 V protection.

Source / changes: Codex circuit using TI TS3A27518E Rev F Fig.5-2, Table 8-1:
https://www.ti.com/lit/ds/symlink/ts3a27518e.pdf
IN1/IN2 high select NO; active-low EN from rotary; 10k pullup.
Channels (Howard 1 Nov 2026, for straight PCB routing):
TCK=4, TDI=6, TMS=3, TDO=5, VREF=2; channel 1 unused.
NO2–NO6: ±6 kV IEC contact rating; not a 5 V DC clamp.
Selection, leakage, power-off and SI need checks.

Function: nine bare test holes for probing the board rails (Chong via Howard, 8 Nov 2026: "is it difficult to fit these voltage header holes as well?"). Holes left to right: +1.2V_DDR, VIO1, +1.8V, DGND, +VDCIN (12 V in), DGND, +3.3V, VIO3, VIO2.
Row below MC1, next to J910/J911; each hole is named on the top and bottom silkscreen. No part fitted (not in BOM or placement).

Source / changes: Opal Kelly BRK8310-A01 schematic p.1 (2022-05-19), test points TP1–TP4, TP7, TP8, TP17–TP19 (footprint Aligni:100, 1.651 mm drill, 2.54 mm pad): https://pins.opalkelly.com/downloads/442/download?category_id=3 ; layout = native import of BRK8310.PcbDoc (9-hole column at 6.18 mm pitch: DGND, +VDCIN, +1.8V, +3.3V, +1.2V_DDR, VIO1–3, DGND; extra DGND points TP5/TP6 not copied).
Changed by Claude: one 1x9 footprint J920 at 3.25 mm pitch in a row; order changed so 12 V sits only beside DGND, VIO1 only beside <=1.8 V rails, and every feed routes without crossings. 0.2 mm feeds from MC1.2, MC2.1/MC2.69, R2 (+1.2V_DDR), the L4 +3.3V plane, J910.17 and the VIO1 run. Measure only. Not checked: probe/clip fit at 3.25 mm pitch. No series resistors (BRK has none): a probe slip between neighbours (e.g. +3.3V/VIO3/VIO2, +1.2V_DDR/VIO1) still connects two rails.

V35 (8 Nov 2026) = V34 + J920 voltage test holes (BRK8310 style, Chong): +1.2V_DDR, +1.8V, +VDC

Sheet: /
File: HDMI_Extension_Board_V35.kicad_sch
**Title: HDMI Extension Board – XEM8310 and USB-C JTAG**
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